QuantVLA: Scale-Calibrated Post-Training Quantization for Vision-Language-Action Models

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[기존 문제]

해결 문제: 상위 언어 모델의 양자화 오차가 확산 트랜스포머(DiT) 기반의 액션 헤드로 전파될 때, 로짓(Logits)의 온도와 잔차 스트림(Residual stream) 에너지가 뒤틀리는 Scale drift 현상이 발생하여 로봇 제어 정책이 붕괴되는 문제를 해결하고자 한다.

- ⇨ **즉, DiT action head VLA에 대한 PTQ 논문**

[Contribution]

- **VLA 시스템 내 DiT 액션 헤드에 성공적으로 적용된 최초의 PTQ 프레임워크**
- **LLM module은 attention projection(Wq, Wk, Wv, Wo) 및 MLP layer 모두 양자화하는 반면, DiT 액션 헤드는 MLP만 양자화하여 attention projection은 부동소수점으로 남기는 선택적 양자화 레이아웃 제안**
- **런타임 오버헤드 없이 역양자화의 스케일을 보정하기 위한 가벼운 어텐션 로짓 보정(ATM) 및 출력 헤드 보정(OHB) 도입**

[Summary]

DiT 기반 액션 헤드에서 발생하는 attention logit 및 출력 drift 문제를 수학적 스케일 보정(ATM, OHB)을 통해 런타임 비용 없이 해결한 VLA 전용 양자화 프레임워크

[Quantization Config]

- **Static/Dynamic: Static PTQ**
- **Weight-only / Weight-Activation: Weight-Activation (W4A8, W4A4)**
- **Granularity: Per-channel (Weights), Per-token (Activations) / Per-head (ATM), Per-layer (OHB)**
- **Quantized Modules: LLM의 모든 Linear layer 및 DiT의 MLP block (단, DiT의 Q, K, V, O Attention projection은 부동소수점 유지)**

[방법]

DuQuant 기반의 블록 직교 회전 및 스무딩을 적용합니다. 캘리브레이션 단계에서 교사 모델과 학생 모델 간의 로짓 표준편차를 맞추는 ATM 스칼라(Head 단위)와 잔차 스트림 에너지를 맞추는 OHB 스칼라(Layer 단위)를 계산한 뒤, 이 스칼라 값들을 추론 시 기존 디양자화(Dequantization) 스케일 값에 곱하여 병합(Fold)합니다.

[Baseline]

pi0.5, GR00T N1.5, OpenVLA (비교 방법론: SmoothQuant, DuQuant)

[Benchmark]

- **환경: LIBERO, Pick-and-Can**
- **평가 Metric: Success rate, Memory savings (GB)**

[Result]

pi0.5 모델에서 기준(97.1%)을 초과하는 평균 성공률 97.6%를 달성하면서도 양자화 모듈의 메모리 사용량을 4.27GB에서 1.28GB로 약 70% 감소

[Insight] 시각-언어 추론부와 DiT 제어부 사이의 인터페이스는 스케일 불일치에 극도로 취약하지만, 매우 가벼운 통계적 스칼라 보정만으로도 전체 연산 구조 변경 없이 안정적인 초저비트 제어가 가능함을 입증.

[Paper Review]

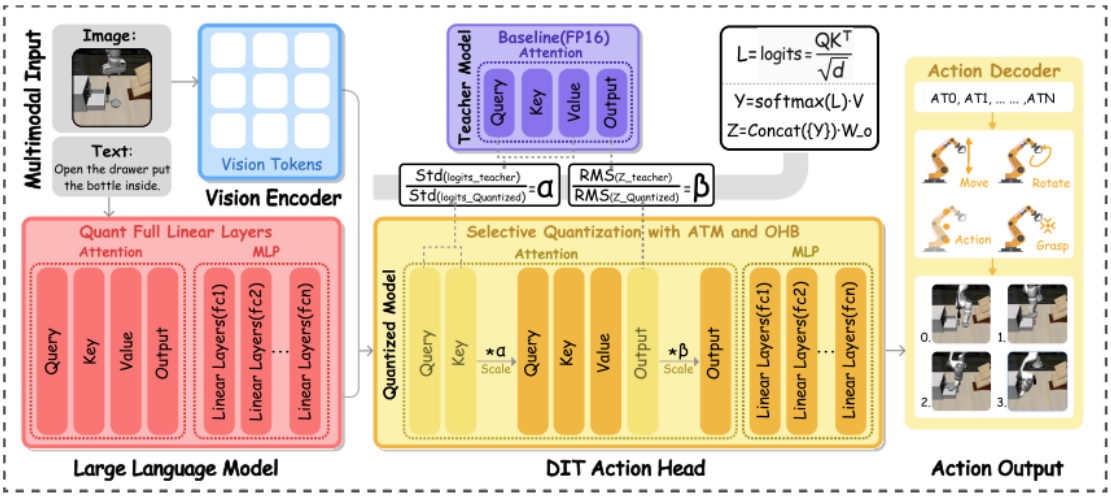

Figure 2. Overview of **QuantVLA** for VLAs with a DiT-based action head. The framework is training-free and preserves the original architecture and operator schedule. It combines: (1) a selective quantization layout that integerizes all linear layers in the LLM and all MLP layers in the DiT while keeping the attention projections Q, K, V, O in floating point; (2) *Attention Temperature Matching* (ATM), a per-head scalar α that aligns teacher–student logits and is folded into dequantization scales; and (3) *Output Head Balancing* (OHB), a per-layer scalar β that matches post-projection energy at the residual interface.

DiT head VLA 구조.

Vision encoder에서 현재 frame에 대한 관측 결과를,
LLM에서 현재 지시사항에 대한 text token을 DiT action head를 통해 행동을 출력

and embedded by a pretrained language backbone. The visual and textual tokens are projected into a shared transformer space, where attention merges perception with the instruction context to form a task-conditioned representation F_{VL}.

The policy head, a Diffusion Transformer, is conditioned on F_{VL}, on robot proprioception, and on a diffusion timestep t. It iteratively updates an action latent according to:

$$x_{t-1} = f_\theta(x_t, F_{\mathrm{VL}}, t). \tag{1}$$

After T refinement steps, the final latent x_0 is decoded into the action. For tokenized policies, the output is a sequence of discrete action tokens. In this formulation, the diffusion transformer denotes the architecture of the policy head. Flow matching [22] denotes the learning objective used to fit f_θ and views the same iterative refinement as a conditional ordinary differential equation (ODE) [34] that trains the network to predict a velocity field transporting x_t toward executable actions.

F_VL을 입력 받아 diffusion denoising을 통해 현재 timestep의 최종 action x0를 생성

VLA Quantization 증폭 과정 증명

VLM에서 생성된 토큰 F_VL을 입력 받아, action head에 넣었을 때 오차가 크게 누적

Building on the deterministic transfer in Appendix C, we now develop a first-order analysis of error propagation to show that the distribution reaching attention is perturbed when DuQuant is applied to the upstream language backbone and the DiT feed-forward blocks. Let X_T denote the teacher input and let $X_Q = X_T + \varepsilon_{\text{up}}$ denote the input under quantization. Even if the attention weights remain in floating point, this perturbation propagates linearly.

$$\begin{aligned} Q_Q &= X_Q W_q = Q_T + \varepsilon_{\text{up}} W_q, \\ K_Q &= X_Q W_k = K_T + \varepsilon_{\text{up}} W_k. \end{aligned} \tag{2}$$

Define the pre-softmax logits L:

$$L_T = \frac{Q_T K_T^\top}{\sqrt{d}}, \qquad L_Q = \frac{Q_Q K_Q^\top}{\sqrt{d}}, \tag{3}$$

and let $\Delta L = L_Q - L_T$. Keeping only first-order terms yields

L_Q term에 Q_q와 K_q를 대입하면 다음과 같이 Logit 변화량이 유도된다,

$$\Delta L \approx \frac{1}{\sqrt{d}} \Big((\varepsilon_{\text{up}} W_q) K_T^\top + Q_T (\varepsilon_{\text{up}} W_k)^\top \Big) + \Delta L_{\text{local}}, \tag{4}$$

where ΔL_{local} aggregates local rounding and scale mismatch from the quantized activations that feed Q and K from the output projection. Let $A = \text{softmax}(L)$ and let $J_{\text{softmax}}(\cdot)$ denote its Jacobian. The attention update satisfies

$$A_Q \approx A_T + J_{\text{softmax}}(L_T)\, \Delta L. \tag{5}$$

A 즉, logit에 softmax를 취한 값에 대해 quantize 된 A(A_q)는

A_T(Full-precision) + Logit에 대한 변화율*변화량 (error) 로 근사할 수 있다.

이제 위 식들을 Value에 곱하고 이를 다시 Output Linear layer에 곱해 최종 출력을 구하면 다음과 같이 quantized – full precision 간 output 오차를 구할 수 있다.

Now include the output head. Write the value path and the quantized output projection as

$$V_Q = X_Q W_v = V_T + \varepsilon_{\text{up}} W_v. \tag{6}$$

The teacher and quantized outputs are

$$O_T = A_T V_T W_{o,T}, \qquad O_Q = A_Q V_Q W_{o,Q}. \tag{7}$$

A first-order expansion around the teacher gives

$$\begin{aligned} \Delta O \approx\ & J_{\text{softmax}}(L_T)\Delta L\, V_T W_{o,T} \\ & + A_T \varepsilon_{\text{up}} W_v W_{o,T} \\ & + A_T V_T \delta W_o \\ & + \Delta O_{\text{local}}. \end{aligned} \tag{8}$$

위에서부터, 상위 모듈(VLM)에 의한 Logit 오차 (1)
Value Weight Quantization 오차 (2)
Output Weight Quantization 오차 (3)
최종 Output 오차 (4) 로 구성된다.

QuantVLA는 오차 (1)을 줄이기 위해, Attention Temperature Matching을, 오차 (3)을 줄이기 위해 Output Head Balancing을 제안한다.

[Attention Temperature Matching: ATM]

$$\alpha_{\text{raw}} = \frac{\text{Std}(L_T)}{\text{Std}(L_Q) + 10^{-6}}. \tag{9}$$

We then confine the correction to a safe range to avoid over-cooling or over-heating the attention distribution

$$\alpha = \text{clip}\big(\alpha_{\text{raw}}, \alpha_{\min}, \alpha_{\max}\big). \tag{10}$$

Next, we apply a neutrality band ε to ignore negligible differences and reduce sensitivity to calibration noise

$$\text{if } \big|\log \alpha\big| < \varepsilon \text{ then } \alpha = 1. \tag{11}$$

Therefore, the quantized logits become

$$L_Q = \frac{L_T}{\alpha}. \tag{12}$$

상위 모듈의 오차로 인해 Q,K가 틀어지고, 이를 보정하기 위해 ATM 적용

단순히, calibration dataset을 가지고, Full-Precision과 Quantized 표준편차를 구하고 이를 alpha로 구한 후 quantized logit에 곱하는 게 끝.

[Output Head Balancing]

We next match the post-projection energy at the residual interface to stabilize the residual injection gain and the operating point of layer normalization. The activation of the output head at the layer l is

$$Z_l = \text{Concat}\{A_{l,h}V_{l,h}\}\, W_{o,l} + b_{o,l}. \tag{13}$$

We measure per-layer energy using RMS for the teacher and the quantized, and directly form a teacher-to-student ratio

$$\beta_{\text{raw}}(l) = \frac{\text{RMS}(Z_{T,l})}{\text{RMS}(Z_{Q,l}) + 10^{-6}} \tag{14}$$

Similarly to ATM, we confine this factor to a safe range and apply a neutrality band

$$\beta(l) = \text{clip}\big(\beta_{\text{raw}}(l), \beta_{\min}, \beta_{\max}\big), \tag{15}$$

$$\text{if } \big|\log \beta(l)\big| < \varepsilon \text{ then } \beta(l) = 1. \tag{16}$$

Finally, we rescale the activation of the output head that enters the residual path

$$Z_Q = \frac{Z_l}{\beta(l)}. \tag{17}$$

OHB도 동일하게 진행

[Experiment]

Model	Precision	Layer Selection	Layer Nums	LIBERO					Memory (GB) (LLM+DiT)
				Spatial	Object	Goal	Long	Avg.	
π0.5	FP16	No Quantization	0	98.5%	99.0%	97.5%	93.5%	97.1%	4.27
	W4A8	LLM	126	98.0%	98.5%	97.5%	92.0%	96.5%	1.58
	W4A8	DiT	126	81.5%	94.5%	71.5%	39.0%	71.6%	3.85
	W4A8	LLM+DiT	252	86.0%	97.5%	71.5%	50.0%	76.3%	1.17
	W4A8	LLM+DiT (MLP)	180	98.0%	97.0%	94.5%	92.0%	95.4%	1.28
GR00T N1.5	FP16	No Quantization	0	92.0%	92.0%	86.0%	76.0%	86.5%	2.02
	W4A8	LLM	84	86.0%	92.0%	80.0%	80.0%	84.5%	1.25
	W4A8	DiT	96	88.0%	80.0%	86.0%	78.0%	83.0%	1.49
	W4A8	LLM+DiT	180	66.0%	70.0%	68.0%	76.0%	70.0%	0.74
	W4A8	LLM+DiT (MLP)	116	90.0%	86.0%	80.0%	74.0%	82.5%	0.91

Table 1. Selective layer-quantization results under the QuantVLA architecture without ATM/OHB calibration for π0.5 and GR00T N1.5 on LIBERO.

LLM + DiT(MLP만) 적용하는 것이 가장 memory와 성능 trade-off에서 최적

Model	Precision	LIBERO					Memory (GB) (LLM+DiT)	Relative Savings
		Spatial	Object	Goal	Long	Avg.		
π0.5	FP16	98.5%	99.0%	97.5%	93.5%	97.1%	4.27	0.0%
+DuQuant(LLM+DiT)	W4A8	86.0%	97.5%	71.5%	50.0%	76.3%	1.17	72.6%
+QuantVLA(LLM)	W4A8	98.5%	99.0%	96.5%	96.5%	**97.6%**	1.58	63.0%
+QuantVLA	W4A8	98.5%	98.0%	98.0%	96.0%	**97.6%**	1.28	70.0%
GR00T N1.5	FP16	92.0%	92.0%	86.0%	76.0%	86.5%	2.02	0.0%
+DuQuant(LLM+DiT)	W4A8	66.0%	70.0%	68.0%	76.0%	70.0%	0.74	63.4%
+QuantVLA(LLM)	W4A8	96.0%	94.0%	92.0%	66.0%	87.0%	1.25	38.1%
+QuantVLA	W4A8	96.0%	92.0%	90.0%	74.0%	**88.0%**	0.91	55.0%

Table 2. Results on LIBERO for different QuantVLA variants on OpenPI π0.5 and GR00T N1.5. The table reports success rates (%) across four LIBERO tasks, memory (GB), and the relative memory savings versus each model's baseline.

단순히 LLM과 DiT에 전부 쓰는 것보다 LLM 전부와 DiT의 MLP만 건드는 것이 최적